

Hydrosaltator amictus Core Statistics

Scientific Name: *Hydrosaltator amictus* (The cloaked hydraulic jumper)

Common Names: Volcanic Spring Tick, Sulfur Spring Tick

Size: 0.3mm | **Total Cell Count:** ~1,400 cells

Lifespan: 1 to 2 active years (can survive centuries in a dormant Tun state)

Cellular Structure

- **Eutelic Core:** The brain, sensory palps, and internal hydraulic valves have a fixed cell count for perfect reflex precision. They cannot heal.
- **Regenerative Shell:** The outer skin, limbs, and pouches can slowly divide to heal and expand.

Thermal Boundaries & Habitat

Habitat: Hydrothermal vents at depths of 2,500m to 4,960m, living in groups disguised as volcano slime.

- **Optimal Active Range:** 50°C to 85°C (122°F to 185°F). Extreme heat powers its symbiotic microbes.
- **The Cryptobiotic Trigger:** Below 40°C (104°F) triggers the tick to enter its dormant Tun state.
- **The Revival Trigger:** Above 60°C (140°F) rehydrates and wakes the dormant tick.
- **The Lethal Limit:** Above 110°C (230°F) destabilizes its pressurized cellular body.

Anatomy & Triple-Symbiosis System

Because the tick completely lacks an internal gut, it relies on a nested symbiotic system:

- **Gelatinicollum vulcanum (Crucible Slime):** A clear, highly elastic slime blanket matrix that anchors micro-partners and builds a protective shield around the raw epidermis.
- **Sulfuribacillus thermophilus (Brimstone Dust):** Deep-layer chemotrophic archaea that harvest volcanic hydrogen sulfide (H₂S) gas and heavy metals, excreting sulfurous ash.
- **Saccharoplasma lentus (Sugar-Silk Protist):** Mid-layer middlemen that consume the archaea's toxic ash byproduct and refine it into glucose and complex lipids.
- **Absorptive Epidermis (~300 cells):** Specialized outer skin that absorbs these sugars directly while pumping out cellular waste to feed the microbes back.

Abilities & Defensive Chemical Profiles

Hydraulic Snap-Jump: Instantly compresses internal hemolymph pressure to snap its 8 limbs together, launching it up to 40mm away (equivalent to a human leaping over 240 feet).

Chemical Component	Conical Pre-Wash Nozzle (Grooming / Slime Soap)	Aerosolizing Blast Vent (Defense / Blinding Cloud)
Hyper-Saline Brine	65%	75%
Liquid Sulfur Solvent	20%	8%
Active Sulfuric/Polythionic Acid	14%	15%
Dissolved Hydrogen Sulfide Gas	1%	2%